# CCA Exam Management System

System Demonstration Guide

Prepared for Panel Presentation

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System URL: http://127.0.0.1:9999

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# 1. System Overview

The **CCA Exam Management System** is a full-stack web application built to automate and streamline the exam scheduling, proctoring, and management process. It replaces manual spreadsheet-based scheduling with an intelligent algorithm that respects real-world constraints.

## Technology Stack

| Layer | Technology |
|---|---|
| Backend | Python 3 / Flask |
| Database | MySQL (via XAMPP) |
| Frontend | HTML5, TailwindCSS, JavaScript |
| Authentication | bcrypt password hashing |
| Email | Flask-Mail (SMTP integration) |
| Export | ReportLab PDF generation |
| Architecture | MVC pattern with modular system/ package |

## Key System Files

| File | Purpose |
|---|---|
| app.py | Main Flask application — all routes (1,277 lines) |
| system/database.py | Database access layer with 50+ functions |
| system/scheduler.py | Automated scheduling algorithm (279 lines) |
| system/exporter.py | PDF export functionality |
| system/logic.py | Authentication & password hashing |
| system/importer.py | CSV import parsing |

# 2. Authentication & Role-Based Access

The system uses **session-based authentication** with bcrypt-hashed passwords. Upon login, users are automatically routed to their role-specific dashboard.

## Login Page

- Clean login form with email and password fields
- Flash message support for errors and success notifications
- Link to student self-registration

## Five User Roles

| Role | Dashboard | Access Level |
|---|---|---|
| Super Admin | /super_admin | Full system control, user management |
| Admin | /dashboard | Exam/room/instructor CRUD, scheduling |
| Proctor | /proctor | View assignments, set availability |
| Secretary | /secretary | Track exam paper collection/return |
| Student | /timetable | View personalized exam timetable |

## Student Registration

- Students select their **institute** and **section** during registration
- Passwords are **bcrypt-hashed** before storage
- Duplicate email addresses are automatically rejected

# 3. Admin Dashboard — The Control Center

The admin dashboard at **/dashboard** serves as the central hub for managing all exam-related data. It provides real-time statistics and quick-access forms organized into tabbed sections.

## Room Management

- Add rooms with: name, capacity, building, floor, institute
- **PWD accessibility flag** — ensures wheelchair-accessible rooms are properly tracked
- **Bulk CSV import** — upload rooms in batch via /import/rooms

## Instructor / Faculty Management

- Add instructors with: name, department, institute, building, PWD flag
- **Subjects taught** field — prevents conflict-of-interest proctoring
- Activate/deactivate toggle — only active instructors are used during scheduling
- When an instructor is added, a **proctor login account** is automatically created
- **Bulk CSV import** — upload instructors via /import/instructors

## Exam Management

- Create exams with: subject name, section, type (Major/Minor), duration, institute
- Exams start with **Pending** status until scheduled
- **Bulk delete** — admin can clear all exams for fresh scheduling

# 4. Institute & Subject Management

## Institutes

Academic institutes (e.g., ICSLIS, IBED, ITHM) are the organizational units of the system. Each institute has a unique **code** and a **default capacity**. Institutes are linked to rooms, instructors, subjects, and exams.

## Subjects

- Full CRUD operations with institute association
- **Common Subject flag** — marks subjects taken across all institutes
- **Content Group Linking** — two subjects from different institutes sharing the same exam content can be linked so the scheduler assigns them the same time slot (prevents exam leaks)
- **CSV Import** — bulk upload subjects with institute mapping

**Why Content Groups Matter:** If MATH 101 in the Engineering institute and MATH 101 in the Business institute share identical exam questions, linking them ensures both exams happen simultaneously — preventing students from sharing answers across time slots.

# 5. The Automated Scheduler (Core Algorithm)

**This is the heart of the system.** The scheduler at /schedule takes all Pending exams and automatically assigns them to time slots, rooms, and proctors while respecting 7+ real-world constraints.

## How It Works

- 1. Admin sets the **exam period** (start and end dates)
- 2. Clicks **Generate Schedule**
- 3. The algorithm processes all Pending exams through the constraint engine
- 4. Results are saved and displayed as a timetable grid

## Scheduling Constraints Enforced

The scheduler respects **7 constraint rules** simultaneously:

| # | Constraint | How It Works |
|---|---|---|
| 1 | Max consecutive proctor hours | Tracks cumulative blocks per proctor per day (configurable, default 8 hrs) |
| 2 | No self-subject proctoring | A proctor will never supervise an exam in their own subject |
| 3 | Max 2 majors per section per day | Prevents student overload — tracked via maj_count[section][date] |
| 4 | Room capacity & availability | Time-interval overlap detection ensures no double-booking |
| 5 | Proctor availability windows | Honors each proctor's declared available dates and times |
| 6 | PWD proctor matched to PWD room | Proctors with disabilities are only assigned to accessible rooms |
| 7 | Travel buffer between buildings | Configurable gap (default 15 min) between assignments in different buildings |

## Content Group Scheduling

Subjects linked as **identical content** are grouped and scheduled at the **same time slot** across all sections — this is critical for preventing exam content leaks between time slots.

## Algorithm Flow

- 1. Retrieve all Pending exams from the database
- 2. Group exams by content group or subject name
- 3. Sort groups: Major exams first, then by name
- 4. For each group, iterate through all date + slot combinations
- 5. Check all 7 constraints for each candidate slot
- 6. If all constraints pass, greedily assign rooms and proctors
- 7. Lock the time slot for this subject (content group consistency)

- 8. If no valid slot found, mark group as Failed/Unscheduled

## Post-Scheduling Actions

- **Email notifications** are automatically sent to assigned proctors via SMTP
- **In-app notifications** are created for all affected users
- Failed exams are highlighted with **specific failure reasons**

# 6. Schedule View & Timetable

## Admin Schedule View

The generated schedule is displayed as an **interactive timetable grid** where columns represent exam dates and rows represent time slots. Each cell shows the subject, section, room, and proctor.

### Conflict Detection

- **Room double-booking** — same room assigned to overlapping times (highlighted in red)
- **Same subject at different times** — potential content leak risk (highlighted in yellow)

### Inline Schedule Editing

- Admin can edit any scheduled exam: change room, proctor, or date/time
- **Real-time conflict checking** via AJAX call to /check_edit_conflict
- Force-save option to override warnings when necessary

### Schedule Reset

The admin can reset the entire schedule, returning all exams to Pending status for re-scheduling.

## Student Timetable (/timetable)

- Students see only **their section's and institute's exams**
- Exams grouped by date with clean cards showing time, subject, room, and proctor
- Read-only access — no editing capabilities

# 7. Proctor Dashboard

Accessible at **/proctor** by proctor-role users. This dashboard provides proctors with everything they need to manage their exam assignments.

| Feature | Description |
|---|---|
| My Assignments | List of all exams they're proctoring, grouped by date with time and room |
| Total Hours | Running total of proctoring hours for the current exam period |
| Set Availability | Select available dates and time windows within the exam period |
| Report Absence | Self-report inability to proctor with reason — automatically notifies admin |

# 8. Secretary Dashboard

The secretary dashboard at **/secretary** tracks exam paper logistics throughout the exam period.

| Action | What It Does |
|---|---|
| Mark Collected | Records which instructor handed in exam papers, with optional notes |
| Mark Returned | Records when graded papers are returned to the department |
| Collection Stats | Live counters showing collected, returned, and pending papers |

**PDF Export:** The secretary can download a collection report as a PDF document.

# 9. Reports & Analytics

## Three Report Categories

**1. Proctor Absence Report**

- Filterable by date range
- Shows proctor name, exam, reason, and who reported the absence

**2. Assignment Report**

- Per-proctor breakdown of exam assignments
- Helps identify workload imbalance across faculty

**3. Scheduling Report**

- Overall statistics: total exams, scheduled vs. unscheduled
- Identifies scheduling bottlenecks and constraint conflicts

## PDF Exports

| Export | Route | Content |
|---|---|---|
| Exam Schedule | /export/schedule.pdf | Full timetable with conflict highlights |
| Absence Report | /export/absence_report.pdf | Proctor absence data + hours summary |
| Collection Report | /export/collection_report.pdf | Paper collection tracking status |

# 10. Super Admin Panel

Accessible at **/super_admin**, restricted to users with the **is_super_admin** flag set.

## Capabilities

- **View all users** across all roles in a single table
- **Edit any user:** change name, email, role, section, institute, and super admin flag
- **Delete users** with protection against self-deletion
- **Reset passwords** — generates a new bcrypt hash for the specified user

# 11. System Settings

The settings page at **/settings** allows admins to configure scheduling parameters and email integration.

| Setting | Default | Purpose |
|---|---|---|
| Day Start | 07:00 | Earliest possible exam slot |
| Day End | 20:00 | Latest possible exam end time |
| Slot Interval | 60 min | Time gap between consecutive slot start times |

| Setting | Default | Purpose |
|---|---|---|
| Max Proctor Hours | 8 hours | Maximum consecutive proctoring hours per day |
| Travel Buffer | 15 min | Required gap between assignments in different buildings |
| SMTP Configuration | — | Host, port, username, password, TLS settings for email |

**Test Email:** Admins can send a test email to verify SMTP configuration is working correctly.

## 12. Notification System

- **In-app notifications** for all user roles
- **Unread badge counter** displayed on every page in the navigation
- **Mark read** — individually or all at once
- **Role-based notifications:** e.g., admin is notified when a proctor reports absence
- **Email notifications:** sent automatically for schedule assignments and changes via SMTP

# 13. Recommended Live Demo Flow (~18 minutes)

| Act | Feature | Duration | Steps |
|---|---|---|---|
| 1 | Login & Routing | 2 min | Open the system, show login page, log in as admin |
| 2 | Data Management | 3 min | Show institutes, subjects, rooms tab, add a room, show instructors |
| 3 | Exam Creation | 2 min | Add new exams, show Pending status |
| 4 | Automated Scheduling | 3 min | Set exam period, generate schedule, show timetable grid, edit an exam |
| 5 | Proctor Experience | 2 min | Log in as proctor, show assignments, set availability, report absence |
| 6 | Secretary Experience | 1 min | Log in as secretary, mark collected/returned |
| 7 | Student Experience | 1 min | Register a student, show personalized timetable |
| 8 | Reports & Export | 2 min | Show all 3 report types, download PDF exports |
| 9 | Super Admin | 1 min | Show user management and password reset |
| 10 | Settings | 1 min | Show configurable parameters and SMTP config |

## Detailed Steps for Act 4 (Core Demo)

This is the most important part of the demonstration — the automated scheduling algorithm:

- 1. Navigate to **Schedule** page (/schedule)
- 2. Set the exam period dates (e.g., August 24-29, 2026)
- 3. Click **Generate Schedule** — watch the algorithm process all pending exams
- 4. Point out the generated timetable: room assignments, proctor assignments
- 5. Verify: no room conflicts, no self-subject proctoring, no overloaded sections
- 6. Click on a scheduled exam — demonstrate the **inline edit** feature
- 7. Change the room — show the **real-time conflict checker** in action
- 8. (Optional) Click **Reset Schedule** to return exams to Pending and re-run

# 14. Key Selling Points for Panel Q&A;

**These are the differentiating features that set this system apart from manual or basic scheduling approaches:**

| # | Feature | Why It Matters |
|---|---|---|
| 1 | Constraint-Aware Scheduling | Not random assignment — respects 7+ real-world rules simultaneously |
| 2 | Content Group Linking | Prevents exam leaks by scheduling identical-content exams at the same time |
| 3 | PWD Accessibility | Matches PWD proctors with PWD-accessible rooms automatically |
| 4 | Travel Buffer Logic | Accounts for physical movement time between buildings |
| 5 | Multi-Role Architecture | 5 distinct user roles with appropriate access levels and dashboards |
| 6 | Real-Time Conflict Detection | AJAX-powered checks when editing schedules — instant feedback |
| 7 | Full Audit Trail | Notifications, absence records, collection logs for accountability |
| 8 | PDF Exports | Professional reports for printing, archival, and distribution |
| 9 | Email Notifications | Automatic SMTP-based alerts for proctor assignments and changes |
| 10 | Bulk Data Import | CSV upload support for rooms, instructors, and subjects |

**To start the system:** Ensure XAMPP/MySQL is running, then execute *python app.py* in the project directory. The application will be available at **http://127.0.0.1:9999**.